

1 Introduction

Each spring, in ecosystems across the temperate zone, ecological communities re-assemble from their period of winter dormancy with the hatching of insects, arrival of migratory birds, germination of seeds and emergence of new leaves and flowers. The phenology—or seasonal timing—of these life cycle events can be quite different, even among closely related organism in a shared habitat (Ovaskainen *et al.*, 2013; An *et al.*, 2022). These phenological differences among taxa have often been overlooked in ecological theory outside of plant-pollinator interactions, but growing evidence suggests that they may play a major role in dictating community interactions and shaping the structure and function of ecological communities (Cleland & Wolkovich, 2024; Blackford *et al.*, 2020; Rudolf, 2019). In particular, experiments demonstrate that the order in which species begin their seasonal growth can alter the outcomes of competitive interactions (Dawson-Glass *et al.*, 2025; Buonaiuto & Wolkovich, 2023), and as such, increasing work has linked phenology to theory in cases where the order of events determines community assembly.

Such phenological differences among species can generate seasonal priority effects; when earlier active species gain a competitive advantage by accessing limiting resources before their competitors emerge or arrive, and modifying the environment that later emerging species experience (Fukami, 2015; Weidlich *et al.*, 2020). Studies in plants (Wainwright *et al.*, 2011; Koerner *et al.*, 2008), amphibians (Rudolf & Singh, 2013), insects (Ekholm *et al.*, 2020; Rasmussen *et al.*, 2014) and birds (Samplonius & Both, 2019) have found that even small phenological priority effects, on the order of just a few days, can substantially alter competitive hierarchies in ways that can persist across several growing seasons (Vaughn & Young, 2015).

The demonstrated importance of seasonal priority effects in mediating interspecific competition has led to growing interest in the role of seasonal priority effects in coexistence (Diez *et al.*, 2020; Blackford *et al.*, 2020). Several existing community assembly models that integrate the timing of species arrive or activity suggest timing is fundamental to coexistence—with several studies proposing a fundamental trade-off between early phenology to gain unrestricted access to finite resources versus later later phenology, which requires competing well under low resource conditions ().

Growing interest in understanding seasonal priority effects also highlights one of the major challenges to the field: phenology is not a static trait. Phenology varies highly across years because of the contribution of the environment in determining it for most, if not all, species. Functionally, phenology is the realized product of an interaction between the environment (cues) and species unique physiology, genetic and developmental attributes (Donohue *et al.*, 2005). With environmental cues changing rapidly in response to anthropogenic climate change, phenological shifts are one of the most pronounced indicators of climate change to date (Scranton & Amarasekare, 2017). However, because the environment is only one part of phenology, species' phenological responses to climate change often differ significantly from each other.

These differences are one of the primary mediators of the tradeoff between phenology and competition and, become amplified in a changing climate. Understanding the degree to which a species responds to envi-

ronmental variation, known as **phenological sensitivity** (Δ phenology/ Δ environmental cue), is key to predicting community assembly in an era of global change. Below we outline why integrating phenological sensitivity into community assembly is critical, and outline a path to develop a more robust way to predict assembly in a rapidly changing world.

2 How phenological differences arise

Environmental cues such as temperature, light and moisture are the primary triggers that control phenology (Forrest & Miller-Rushing, 2010). Plants and animals may relay on different cues (Lang *et al.*, 2025), and many species respond to multiple aspects of cues. For example, considering temperature cues, many temperate plants require both exposure to cool winter temperature (0-4C, hereafter: winter chilling) and warm spring temperatures to initiate phenological activity (Ettinger *et al.*, 2020). These sensitivities are the unique product of species physiology, genetics and development. What this means is that phenology itself is a response to the environment, so that *realized phenology differences* between species will vary substantially depending on the environmental conditions.

Decades of seed germination studies provide reliable examples of how realized phenology varies across species $\times$ environment (). We illustrate this point by re-analyzing seed phenology data from Buonaiuto (2023), see extended Methods for details), in which 11 co-occurring species were germinated under varying temperature conditions. Even species from a common environment have very different sensitivities to temperature cues (Figure 1a). As seen in our comparison of two species, *Persicaria virginiana* and *Eurybia divaricata*—a pair of woodland herbs that frequently compete in Eastern Temperate Forests—whichever species is first depends on environmental conditions, and switches with increasing cue levels (Figure 1b,c. in this case, chilling duration). This illustrates that the strength of seasonal priority effect is highly dependent on the environment, yet this is not how most coexistence models approach phenology.

3 Recent work integrating phenology into assembly models

Growing interest in how phenology may structure communities has led to increased modeling efforts, with most approaches centering on the relationship between timing and competition (Rudolf, 2019; Wolkovich & Donahue, 2021; Levine *et al.*, 2022). Building off experimental (Godoy & Levine, 2014) and observational (Wolkovich, 2013) evidence that earlier species appear to perform better, models have generally built in trade-off where species with a timing advantage also experience greater competition. Models have done this through an explicit connection of phenological differences to parameters controlling the strength of competition (Rudolf, 2019), or implicitly by providing a fitness advantage to earlier species that then must be offset in communities where species also vary in parameters related to R^* (Wolkovich & Donahue, 2021; Levine *et al.*, 2022).

Models providing a fitness advantage to earlier species have all recognized the importance of environmental variability in communities structured in part by phenological differences, generally including interannual variation in the environment that in turn drives differences in phenology. Many models—focused often on grasslands and water competition—have assigned phenology as equivalent to an intrinsic species-level timing (or season length strategy). Year to year variation in the environment then alters absolute differences between species, which then consistently co-varies with resource competition (Godoy & Levine, 2014; Levine *et al.*, 2022). In these approaches the relative orders between species remain fixed, but other models allow species relative orders and timings to vary year-to-year assuming the underlying relationship between pairwise differences in timings and interspecific-competition is known—and static. Each of these approaches in turn thus fixes one aspect of environment $\times$ phenology $\times$ competition relationship (i.e., either through an explicit fixed relationship between phenology and competition or fixing the order of species but allowing the strength of competition to vary). This is understandable simplification, but one that may make forecasting community shifts with climate change challenging, as the environment becomes more variable.

Community assembly models including phenology have all included environmental variability (Rudolf, 2019; Levine *et al.*, 2022) assuming an underlying stationary environment (one that varies year-to-year but with an underlying constant distribution), but climate change represents a clearly non-stationary environment. Temperature, for example, trends higher in its mean over time, with potential additional changes to the variance (Canadell *et al.*, 2021). Because environmental variability in assembly models structures what species combinations are stable, shifts in the environment with anthropogenic climate change are likely to reshape communities, but forecasting them requires accurately capturing how the environment affects phenology and competition together in the simplest—but still robust—form. We argue this means capturing how dynamically the environment can determine both absolute and relative timings of species, and the how those timings directly impact co-existence. While current modeling approaches to date have simplified one aspect of this problem, we expect that it can be tractably addressed through new models that explicitly allow these dynamics in both inter- and inter-annual time, and connect to the underlying physiology of phenology.

4 Climate Change, Phenology and Coexistence

Leveraging community assembly models to address how shifting phenology in a non-stationary climate of global warming may affect coexistence requires more flexible models that can encapsulate the strong inter-dependence between environmental variability, phenological variability and variable competition strength. Integrating environmental variability through modeling to its effects on phenology and competition highlights that climate change will disrupt the stability of trait-based, community trade-offs.

We extended a traditional assembly model to include phenological sensitivity in its simplest form—two species with differential sensitivity to one environmental cue. After 1000 runs of 300-year duration under two different climate normals (See Extended Method for details) we found that no species pairs that coexisted

under one climate condition through mechanism phenology/competition trade-offs could continue to coexist through this mechanisms in the other (Figure 2a, see circled points for example). This is because the relative phenological differences between species that structures the tradeoff are dependent on the interaction between each species' phenological sensitivity and the environment itself, and therefore, in a novel environment the relationship between phenological sensitivity differences and realized phenological differences shift (Figure 2b). Thus it follows that for any species that currently coexist due to the tradeoff between relative phenology and intrinsic competitive ability, climate change will disrupt this coexistence mechanism.

Our modeling extension reveal an additional, critical point for improving the predictive power of coexistence model with climate change. While the tradeoff between realized phenological differences and competitive differences is stable across a shifting environment (i.e., the same magnitude if phenological differences are required to offset the same level of differences in competitive ability), it is the tradeoff between phenological sensitivity differences and competition differences that change in a novel environment (Figure 2a,c). Phenological sensitivity is a trait that can be readily measured for real species, providing a link between assembly theory and observed ecological changes. Yet, effectively leveraging this trade-off between phenological sensitivity and competitive ability to model community assembly under future climates will require the continued development of both novel experimental approaches and more-biologically realistic modeling frameworks, but perhaps most importantly, increased integration between the two.

5 Paths Forward

On the experimental side, it is critical studies that quantify species-level phenological sensitivity to multiple environmental cues continue to be developed, and expand the scope taxa and life-forms they include. Open questions in basic eco-physiology, like the extent of local adaptation in phenological sensitivity and plasticity in competitive traits that are already being explored (e.g. (Lipowsky *et al.*, 2015; Liang & Wu, 2021) should be contextualized in the context of ecological forecasting. Estimates of species-level phenological sensitivity, and quantification of variation of these estimates at multiple scales (i.e., population, individual) would allow researchers to parameterize community assembly models for real species, helping to increase the practical application of these models and facilitate the development of predictive tools to aid ecological practitioners.

Most models of ecological competition that include phenological parameters, are developed to encapsulate competition between two annual plants (). However herbaceous perennials and woody plants are the dominant functional types in many ecological systems, and as such phenology-mediated competition occurs between individuals both within- (i.e., seedling vs. seedling or mature plant vs mature plant) and across- (i.e, mature plant vs. seedling) life stages. While studies in woody plants suggest separate phenological phases can respond quite differently to the same environmental cues (Buonaiuto & Wolkovich, 2021) and that juveniles and adults express substantially different phenological patterns in the same environments (Vitasse, 2013), changes in phenological sensitivity to multiple cues across multiple life stages relevant to community assembly

(i.e., seed germination vs vegetative cues) have not been widely explored across taxa. To help modelers can construct more biologically realistic community assembly models that include multiple kinds of growth forms and life stages interacting, experimentalists could design studies that explore how environmental variability influences phenological differences across life-stages and mediates both intra- and inter-specific competition within and across these stages.

On the modeling side, researchers should strive to integrate the more complex cue responses to assembly models. For example, it is well supported that phenology is determined to multiple, interacting environmental cues including multiple aspects of temperature (i.e., forcing and chilling), light (daylength or irradiance) and moisture (Baskin & Baskin, 2014; Forrest & Miller-Rushing, 2010). Selection for interacting cues is generally thought to be driven by another component likely critical for better predictions—environmental risk. Complex cues are expected to reduce risk but also result in a more conservative phenological strategy—for examples in spring phenology, reliance on winter chilling and/or photoperiod reduce the risk of frost damage in unstable spring environments, but generally results a later phenological transitions (Zohner *et al.*, 2017). Ecological models generally subsume such risks into species traits (e.g., increasing maintenance costs), which misses the reality that such risks are also product of timing x environment. A changing risk landscape associated with a changing environment may be the underlying driver of observed correlations between the timing and intensity component of life cycle transitions (i.e., germination phenology and total germination percent of a seed cohort Baskin & Baskin, 2014). Because timing and fraction usually covary in nature, building models that can isolate the two trade-off axes of coexistence, would allow offer to assess the relative importance of these within- and across season trade-offs.

Another dimension of risk and phenology that could help generate more predictive assemble models is the to understand how another aspect of phenology that is sensitive to environmental cues—the duration of a phenological phases—mitigates or exacerbates environmental risk. Environmental cues not only advance or delay the start of a phenological transition, but may also influence the rate of the transition. More protracted phenological phases (i.e., if the germination of a full seed cohort takes longer) associated with unfavorable conditions may serve as a form of within-season bet-hedging in which an untimely mortality event would have a lower fitness cost to a cohort of seed that started germinating early but slowly (at the cohort level) than one that started early and finished fast. Introducing biological realistic mortality events into community assembly models that represent import selective forces like late-season frosts or early droughts, offers a straightforward approach to assess the importance of the often overlooked aspect of phenology.

The modeling framework we present in this study has the flexibility to be further extended to incorporate many of the ecological aspects we discussed above (multiple cues, environmental risk, within- and across-season bet-hedging). But to make these predictions meaningful for the moment, modelers and physiologists must continue to work together to further develop models with biologically accurate physiological processes, and experiments that are readily translatable into model parameterizations and development. In a rapidly changing world, these kind of cross-disciplinary collaborations are critical to translate ecology theory into practice and advance a more predictive ecology than is better equipped to inform ecosystem conservation

173 and management.

174 **Figures**

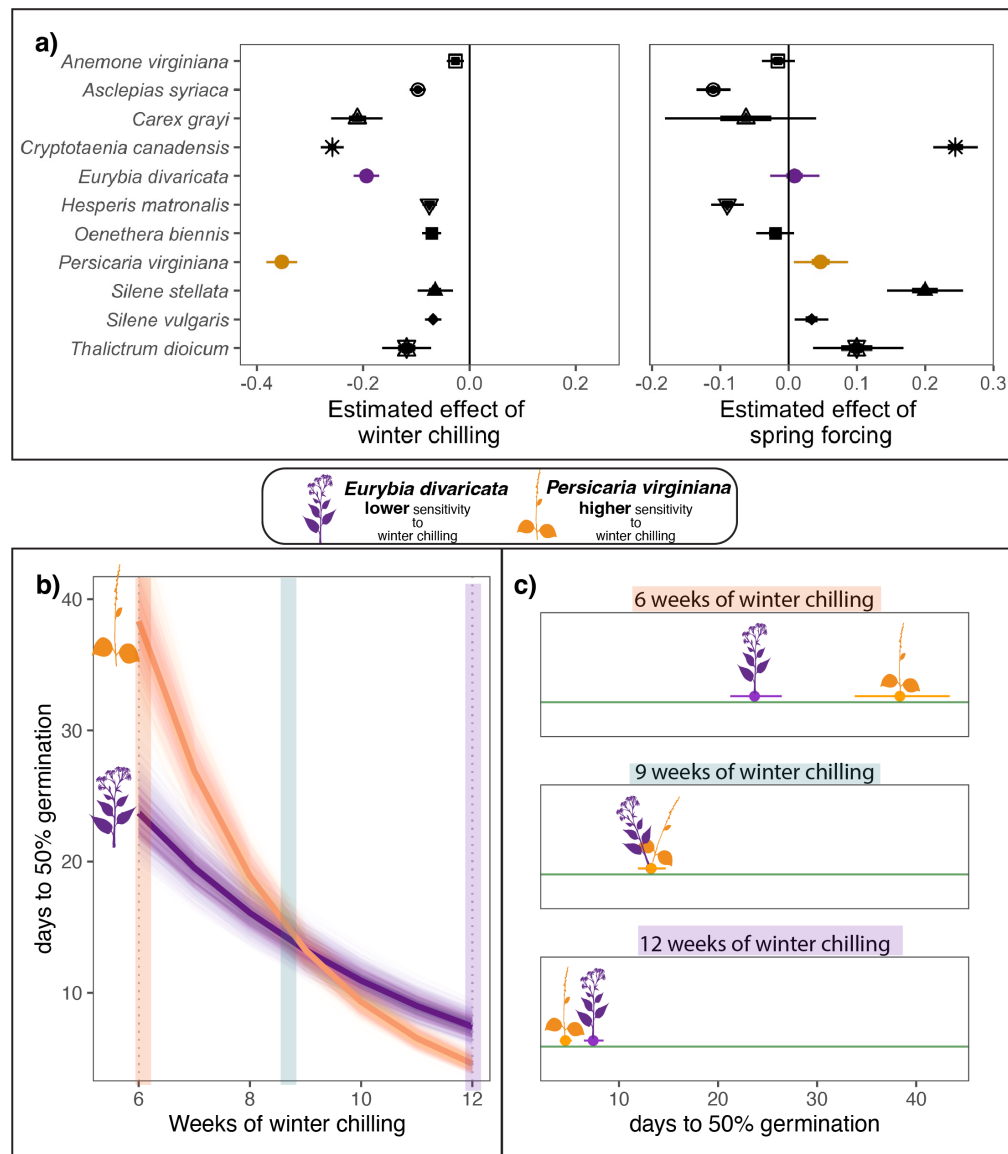


Figure 1: **Co-occur species can differ substantially in their phenological sensitivity (Δ phenological event/ Δ climate) to the environment.** Panel a) estimates the phenological sensitivity of germination timing to winter chilling and spring forcing for eleven herbaceous species of eastern North America. Points represent the mean estimate and thick and thin bars represent the 50 and 95% uncertainty intervals. In b) and c) we highlight the phenological sensitivity to winter chilling of two species *Eurybia divaricata* and *Persicaria virginiana* that are commonly observed in competition. b) Depicts the sensitivity of each species to winter chilling, with thick lines representing the mean estimate and lighter lines representing modeled uncertainty. Panel c) illustrates how the species' different sensitivities manifest in contrasting realized germination differences under different climate conditions. Point represent the predicted mean number of days to 50% germination and bars the 95% uncertainty intervals

References

- An, S., Chen, X., Shen, M., Zhang, X., Lang, W. & Liu, G. (2022). Increasing interspecific difference of alpine herb phenology on the eastern qinghai-tibet plateau. *Frontiers in Plant Science*, Volume 13 - 2022.

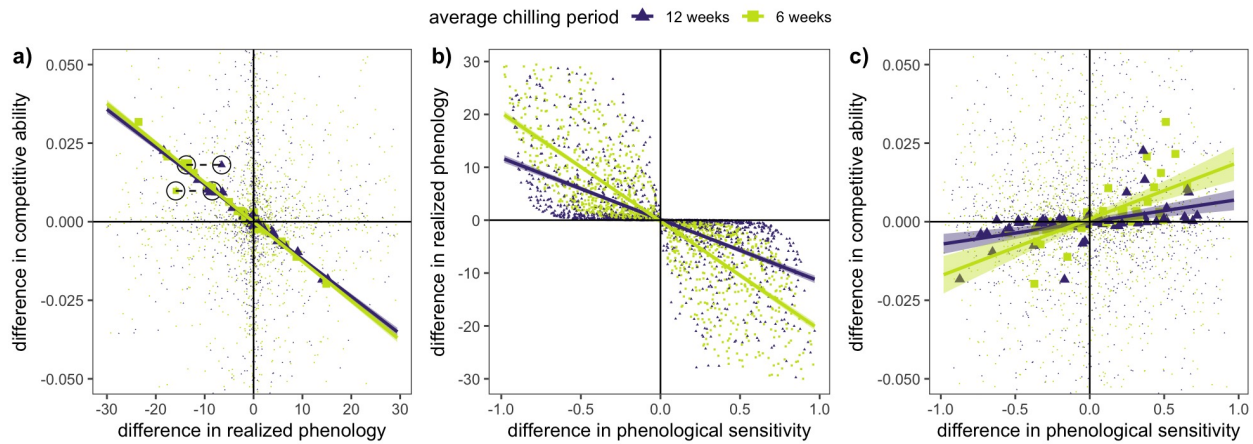


Figure 2: Climate change will alter the relationship competitive ability, phenological sensitivity, realized phenology and coexistence. Panel a) depicts the trade-off between realized phenological differences and competitive differences. While the tradeoff relationship between realized phenology and competition does not shift with climate change, species that coexisted under one climate scenario will not coexist if climate conditions shift, as highlighted by the two sets species pairs circled in panel a). These changes occur because the relationship between phenological sensitivity differences and realized phenology differences, depicted in panel b), are dependent on the environment, and a change in chilling conditions alters this relationship. Panel c) depicts the trade-off between phenological sensitivity differences and competitive differences under two climate scenarios (average of 12 and 6 weeks of chilling). With climate change, the trade-off between phenological sensitivity and competition does shift because less chilling amplifies the realized phenological differences among species.

- Baskin, C. & Baskin, J. (2014). *Seeds: Ecology, Biogeography, and Evolution of Dormancy and Germination*. Elsevier Inc.
- Blackford, C., Germain, R. M. & Gilbert, B. (2020). Species differences in phenology shape coexistence. *The American Naturalist*, 195, E168–E180.
- Buonaiuto, D.M., . W. E. (2023). Germination phenology and seedling competition between two forest herb species of the eastern united states in controlled environments. *Knowledge Network for Biocomplexity*, doi:10.5063/F1BZ64HV.
- Buonaiuto, D. M. & Wolkovich, E. M. (2021). Differences between flower and leaf phenological responses to environmental variation drive shifts in spring phenological sequences of temperate woody plants. *Journal of Ecology*, 109, 2922–2933.
- Buonaiuto, D. M. & Wolkovich, E. M. (2023). Contrasting responses to climate variability generate seasonal priority effects between native and invasive forest herbs. *Journal of Ecology*, 111, 1711–1721.
- Canadell, J., Monteiro, P., Costa, M., Cotrim da Cunha, L., Cox, P., Eliseev, A., Henson, S., Ishii, M., Jaccard, S., Koven, C., Lohila, A., Patra, P., Piao, S., Rogelj, J., Syampungani, S., Zaehle, S. & Zickfeld,

- K. (2021). *Climate Change 2021: The Physical Science Basis. Contribution of Working Group I to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change*. Cambridge University Press, New York, NY.
- Cleland, E. E. & Wolkovich, E. (2024). Effects of phenology on plant community assembly and structure. *Annual Review of Ecology, Evolution, and Systematics*, 55, 471–492.
- Dawson-Glass, E., Schiafo, R., Kuebbing, S. E. & Stuble, K. L. (2025). Warming-induced changes in seasonal priority effects drive shifts in community composition. *Ecology*, 106, e4504.
- Diez, J., Schlauch, J., DuCharme, E., Collings, J. & Erskine, S. (????). Germination phenology alters species coexistence outcomes. *Journal of Ecology*, n/a.
- Donohue, K., DORN, L., Griffith, C., Kim, E., Anna, A., Polisetty, C. R. & Johanna, S. (2005). The evolutionary ecology of seed germination of *arabidopsis thaliana*: Variable natural selection on germination timing. *Evolution*, 59, 758–770.
- Ekholm, A., Tack, A. J. M., Pulkkinen, P. & Roslin, T. (2020). Host plant phenology, insect outbreaks and herbivore communities – the importance of timing. *Journal of Animal Ecology*, 89, 829–841.
- Ettinger, A. K., Chamberlain, C. J., Morales-Castilla, I., Buonaiuto, D. M., Flynn, D. F. B., Savas, T., Samaha, J. A. & Wolkovich, E. M. (2020). Winter temperatures predominate in spring phenological responses to warming. *Nature Climate Change*, 10, 1137–1142.
- Forrest, J. & Miller-Rushing, A. J. (2010). Toward a synthetic understanding of the role of phenology in ecology and evolution. *Philosophical Transactions of the Royal Society B: Biological Sciences*, 365, 3101–3112.
- Fukami, T. (2015). Historical contingency in community assembly: Integrating niches, species pools, and priority effects. *Annual Review of Ecology, Evolution, and Systematics*, 46, 1–23.
- Godoy, O. & Levine, J. M. (2014). Phenology effects on invasion success: insights from coupling field experiments to coexistence theory. *Ecology*, 95, 726–736.
- Koerner, C., Stoecklin, J., Reuther-Thiebaud, L. & Pelaez-Riedl, S. (2008). Small differences in arrival time influence composition and productivity of plant communities. *New Phytologist*, 177, 698–705.
- Lang, W., Zhang, Y., Li, X., Meng, F., Liu, Q., Wang, K., Xu, H., Chen, A., Peñuelas, J., Janssens, I. A. & Piao, S. (2025). Phenological divergence between plants and animals under climate change. *Nat Ecol Evol*, 9, 261–272.
- Levine, J. I., Levine, J. M., Gibbs, T. & Pacala, S. W. (2022). Competition for water and species coexistence in phenologically structured annual plant communities. *Ecology Letters*, 25, 1110–1125.
- Liang, L. & Wu, J. (2021). An empirical method to account for climatic adaptation in plant phenology models. *International Journal of Biometeorology*, 65, 1953–1966.

- Lipowsky, A., Roscher, C., Schumacher, J., Michalski, S. G., Gubsch, M., Buchmann, N., Schulze, E.-D. & Schmid, B. (2015). Plasticity of functional traits of forb species in response to biodiversity. *Perspectives in Plant Ecology, Evolution and Systematics*, 17, 66–77.
- Ovaskainen, O., Skorokhodova, S., Yakovleva, M., Sukhov, A., Kutenkov, A., Kutenkova, N., Shcherbakov, A., Meyke, E. & del Mar Delgado, M. (2013). Community-level phenological response to climate change. *Proceedings of the National Academy of Sciences*, 110, 13434–13439.
- Rasmussen, N. L., Van Allen, B. G. & Rudolf, V. H. W. (2014). Linking phenological shifts to species interactions through size-mediated priority effects. *Journal of Animal Ecology*, 83, 1206–1215.
- Rudolf, V. H. W. (2019). The role of seasonal timing and phenological shifts for species coexistence. *Ecology Letters*.
- Rudolf, V. H. W. & Singh, M. (2013). Disentangling climate change effects on species interactions: effects of temperature, phenological shifts, and body size. *Oecologia*, 173, 1043–1052.
- Samplonius, J. M. & Both, C. (2019). Climate change may affect fatal competition between two bird species. *Current Biology*, 29, 327–331.e2.
- Scranton, K. & Amarasekare, P. (2017). Predicting phenological shifts in a changing climate. *Proceedings of the National Academy of Sciences*, 114, 13212–13217.
- Vaughn, K. J. & Young, T. P. (2015). Short-term priority over exotic annuals increases the initial density and longer-term cover of native perennial grasses. *Ecological Applications*, 25, 791–799.
- Vitasse, Y. (2013). Ontogenic changes rather than difference in temperature cause understory trees to leaf out earlier. *New Phytologist*, 198, 149–155.
- Wainwright, C. E., Wolkovich, E. M. & Cleland, E. E. (2011). Seasonal priority effects: implications for invasion and restoration in a semi-arid system. *Journal of Applied Ecology*, 49, 234–241.
- Weidlich, E. W. A., Nelson, C. R., Maron, J. L., Callaway, R. M., Delory, B. M. & Temperton, V. M. (2020). Priority effects and ecological restoration. *Restoration Ecology*, n/a, e13317.
- Wolkovich, E. & Donahue, M. J. (2021). How phenological tracking shapes species and communities in non-stationary environments. *Biological Reviews*, 96, 2810–2827.
- Wolkovich, E. M. (2013). Flowering phenology of native and exotic species across five sites, <http://knb.ecoinformatics.org/knb/metacat/wolkovich.41/knb>.
- Zohner, C. M., Benito, B. M., Fridley, J. D., Svenning, J.-C. & Renner, S. S. (2017). Spring predictability explains different leaf-out strategies in the woody floras of north america, europe and east asia. *Ecology Letters*, 20, 452–460.